Drew University - CSCI 260 - Fall 2024
HW5 - LC-3 Assembly Language Part II

HW Info.: Due on 11/08/2024 (F) by 11:59pm, 30 points
Submission: Submit as a single PDF file on Moodle.
Note: Please follow Drew University's Standard of Academic Integrity and refer to the course syllabus for late submission policy.

1. (15 points) Write an LC-3 assembly language program that performs division. If we manually set R0 to be the divided and R1 to be the divisor, then use R2 to store the quotient and R3 for the remainder. For example, R0=40 (dividend), R1=3 (divisor), then since R0=13 $\times$ R1+1, R2=13 (quotient) and R3=1 (remainder).

   [**Hint**: Try to replicate how multiplication is done.]

2. (15 points) The *Euclidean Algorithm* is used to find the *Greatest Common Divisor* (GCD) of two positive numbers; i.e., $GCD(10, 15) = 5$, as $10 = 2 \times 5$ and $15 = 3 \times 5$, where 5 is a (and only in this case) common divisor between 10 and 10; and since 5 is the largest such divisor, it is the GCD.

   Below are the steps of the algorithm for computing GCD:

   (i) We put the larger number into R1 and the other number into R2.

   (ii) Keep subtracting R2 from R1, until R1 contains a smaller value than R2 does.

   (iii) From Step (ii), if R1 contains a positive number, we exchange values in R1 and R2, and repeat Step (ii).

   (iv) From Step (ii), if R1 contains a zero, we stop the program. What's in R2 is now the GCD.

   Write an LC-3 assembly language program that implements the *Euclidean Algorithm*.

   [*Hint*: Might have additional steps in between; utilize CC for comparisons.]

   Example: GCD(624, 36) [i.e., R1 = 624 and R2 = 36] = GCD(588, 36) = GCD(552, 36) = ...
   = GCD(48, 36) = GCD(12, 36) = GCD(36, 12) = GCD(24, 12) = GCD(12, 12) = GCD(0, 12) = 12.